

INFOMSCIP 2026-2027

Assignment 3: Experimental user study

General comments: Remember the general statements from previous assignments about carefulness and taking proper time. For this one, also consider that there is a need to collect data from users. Make a good time planning at the beginning and consider some buffer at the end in case there are any unforeseeable problems coming up.

General rules: Send your final group report to me by email (a.elali@uu.nl) as a PDF.

- Download the zip package 'acmart-primary.zip' from the Assignments page on Brightspace, and use sample-sigconf.tex. Check 'latex_template_readme.pdf' for instructions.
- For the PDF file, use the name INFOMSCIP-X-Y.pdf where X is the number of the assignment and Y is your group number. Put all group members' names and student IDs at the beginning of the PDF and report here the time spent by each group member separately. You can do this in the author fields by presenting your name as e.g., *Abdo El Ali - 7##### - 10 hours*.

Use of Generative AI / LLMs for writing is not allowed: Please avoid using LLMs for writing in this (or future) assignments. Three reasons: (1) while LLM-generated language may look readable and plausible, the resulting work is often shallow, generic, or insufficiently grounded, which may be apparent during assessment; (2) while such tools can be useful for many tasks, relying on them for assignments like this can reduce your own learning and development. This applies not only to this course, but across your Master's program, and ultimately defeats the purpose of completing Master's-level work; and (3) submissions may be screened using AI-text detection tools, including Pangram, GPTZero, etc. Detector results are indicators and may lead to further review by the examination committee. You may however use AI tools for vibe coding your prototype(s).

Important note: Make sure to read **the whole assignment** carefully before starting it. Step 3 will require you to make some compromises that might have an impact on your decisions in Step 1. This is not bad planning from my side, but a general aspect of research: you always have to think ahead and plan your whole approach beforehand to avoid running into problems at later steps. Make sure each member has read (corrected if necessary) and approved the final version.

Deadline for this assignment: Submit the final version of the PDF before **Thursday, October 8, 2026, 11:00**. Deadlines are strict. Each additional day after the deadline will result in a grade deduction of 1.0.

General remarks: This assignment is about *experimental user studies*. You will specify related research questions and/or hypotheses, design the experiments, execute them with a few people, carry out a user experience survey/interview and analyze the data (to some degree). The topic will be related to typing on mobile phones. We assume that each member of your group has access to a smartphone that you can use for these tests.

To deliver: The result of this assignment should be delivered as a PDF file. Start this file with a list of your team members. Before handing it in, add a rough estimate of hours spent on this assignment for each team member. In the acknowledgements, give a short description of special tasks that each person has done (unless you split all tasks equally or did all work together). One sentence is enough for this; no details required unless you deem them necessary for your grading. You can use the acknowledgements environment (*acks*) for this at the end of the file.

First step: *Specification of the research.* Before we do research, we need to motivate it, introduce our goals, and specify our research correctly. The latter is usually done by introducing a research question and a hypothesis (but other options exist).

The study involves both quantitative and qualitative (it may also involve quantitative subjective) user study design, experimentation and reporting. Thus, all steps will have two sub-tasks, one to specify/design/evaluate the quantitative aspect, another to evaluate a “softer” characteristic, such as “game play experience” or “usability”.

Here, we want to study typing on mobile phones with a touch screen and a so-called *soft keyboard*, i.e., a set of on-screen keys operated via a touch screen. There are plenty of ways to implement such a keyboard and what option you chose will have an impact on typing behavior, performance, and experience. Because the focus here is on doing an empirical study, we are not looking into new creative ways to implement on-screen typing, but want to study existing approaches. For the sake of this assignment, you can also skip the literature study that should usually be done before you start your actual research. That being said, you are more than welcome to review the relevant literature and back up your research (from questions to design and experimentation) with relevant literature.

Sit together with your group and discuss possible options. Check first what kind of phones you have available, and see how their keyboards are implemented and what parts of the implementation are interesting and suitable to study with an empirical experiment. This can be on a rather “high level”. For example, different manufacturers might do things a bit different (does it, e.g., make sense to compare

Android vs. iOS keyboards?), or offer alternative ways for text input (e.g., proposing closest words to the typed incomplete word or words that frequently appeared after the last typed word in addition to normal typing). Or you can study a more “low level” feature. For example, different types of feedback are often provided when typing on a soft keyboard, including changing visuals, vibrations, or sounds when you hit a key. The latter two can often be turned off, so you could evaluate their impact on typing performance. Check what options the device(s) that you have available offer and decide on an interesting and relevant aspect that you want to study. Also, make sure that you actually *can* study this aspect, i.e., that the results of your experiment will indeed allow you to draw conclusions (which requires you to already think about what you will be measuring and how), and consider pragmatic decisions (will you have devices available for testing, etc.). Note that there are also typing apps out there that allow you to record things such as typing speed. If you know one or are eager to do a little research to find one, you are welcome to use it, too. Do keep in mind however that some of these existing solutions may come at the cost of being open and transparent.

Once you decided on a problem to study (i.e., your research aim or general goal), specify it concretely (i.e., phrase a research question and/or a hypothesis for each sub-task). Make sure to consider the characteristics that we discussed in the lecture when phrasing your research question. Keep in mind that it is okay and sometimes even essential that your research question narrows down the scope of your research problem.

It is up to you to decide what kind of a cause-effect relationship to investigate. Make sure that the same setting may also be used for analyzing user experience via an interview or user experience questionnaire. To have fun while doing your project, you may design a pairwise simultaneous typing competitions and exchange the screenshots with each other and hence collect data.

To deliver: After listing all your team members at the beginning of the PDF, address the things mentioned above. Be precise and to the point. No lengthy explanations with issues of minor relevance.

Use the following structure (but slight deviations are acceptable if they make sense and serve your purposes):

- Research aim / general goal / research problem: ...

Write one sentence or phrase to introduce it. Then add some text to shortly explain it; 1-2 sentences could be enough if you phrase it nicely, but can be longer, up to 3-4 sentences.

- Research questions: ...

one for the quantitative task, one for the user experience task
Phrase your questions. If necessary, further explain them.

- Hypotheses: ...

Not all research needs an hypothesis, but for empirical research it is often helpful (and if you do a statistical analysis it might be required). Thus, phrase one, and add some text explaining it (the text should illustrate why you make this assumption about the outcome of your experiment).

Feel free to add any other text that you deem relevant, but avoid adding unnecessary information or issues of minor relevance (again, be precise and to the point). It is totally okay, and might even be better, not to use full text but phrasings and a bullet point structure.

Some related comments and grading criteria: The major issues and thus criteria for the grading are: Is the problem relevant and interesting? How well are the research questions, hypotheses, and/or other type of specification formulated. Is it specific? Can it be answered, verified, or resolved with an empirical study? Grading will also depend on how well you are “to the point”, that is, both missing relevant information (i.e. low recall) as well as providing too much irrelevant content (i.e. low precision) could both have a negative impact on your grade.

Second step: Experiment design. Now, you need to set up and clearly define the design of your experiment. Identify the variables (including their type, e.g., independent and dependent, control, random, etc.) and measures used, as well as the following things:

- Subjects: who, number, (statistics on demographics) ...
- Environment

- Equipment / material used
- Other contexts
- Tasks to perform, instructions given, ...
- Within/between-subjects design
- Order (for within-subjects design) and other mappings (for both)
- Other considerations, if any

Notice that this list might not be complete and some of these items might not be needed for your test. There might also be things that we discussed in the lecture that are not listed here. And there might be relevant other aspects that we have not discussed yet or will not address at all, because there are too many of them, and some might only apply in special cases, such as your concrete study. See also the related remarks about grading below. Also, you will need to make some compromises in the setup because this is a short duration assignment, not a full-fledged research project (see the information in Step 3).

For the qualitative (or subjective quantitative) user experience analysis task, an obvious way to test for more “softer goals” such as usability, workload, engagement, game play experience (if you opt for a competition), and so on is to use a standard questionnaire that evaluates this particular aspect (if such a standardized questionnaire exists). In any case, make sure that you explain and justify your choice. That is, shortly state why this questionnaire is good for your research. For example, you can shortly illustrate certain characteristics of it and why these make this questionnaire well suited for your goals.

Also think about if this single questionnaire is sufficient to answer your user experience related research question or if other means are helpful to make a stronger conclusion. Likewise, you could consider complementing your questionnaire with some qualitative statements gathered via an interview.

To deliver: Add the study design to your PDF document. Make very sure that all information is there and clearly and nicely represented. We recommend using the layout to structure it in a way that key issues are easy to spot (e.g., don’t write full text but use bullet lists, font style, sub-sections etc.). We often see reviews of submitted papers where the reviewer complains about missing information that is actually there, but gets just overlooked because it is badly represented. It is your job as authors to make sure that the reader gets everything, not the readers’.

Some related comments and grading criteria: Clarity and completeness are the most important grading criteria for this step. If you miss specifying things that have been discussed in the lecture and are of relevance for your experiment, you will get a deduction in your grade. If you miss specifying relevant things that have NOT been discussed by the distribution of this assignment, you will not get punished, but you can get extra credit if you consider them. Again, feel free to add any text that you deem relevant, but avoid adding unnecessary information or issues of minor relevance. The comments above about “being to the point” and presenting your work accurately apply here as well.

Third step: *The actual experiment.* In the second step, you (hopefully) specified how many subjects you need for your experiment. Given the time and scope of this assignment, this will very likely not be feasible. But the purpose of this assignment is not to do a full research project but to learn how to do empirical research, so we can make a compromise here. For the sake of this experiment, assume you want to make a dry-run with some people before the actual test to see if your setup and everything else works as planned. (Note: We recommend to do this with at least one to two person(s) before any real experiment, e.g., in your master thesis, as well). For this assignment, other group members can mutually serve as test subjects. We will use the following pairs of groups: 1&2, 3&4, 5&6, 7&8, 9&10, 11&12. That is, each of these groups should run a pilot test with 3-4 group members from the other one.

Because we use existing implementations, we are also a bit limited in recording the data. For example, the time may be too short to implement an automatic tracking and analysis of typing actions. Likewise, it may not be feasible to implement automatic time logging. Yet, you will probably be able to track typing errors by either observing the participants or having them type in a document that you can analyze later (but notice that you cannot analyze, e.g., the number of corrections in such a case). You are allowed to make compromises (e.g., it might be okay to measure the timing with a separate stop watch; for some time measurements this might give you a sufficient approximation, for others not). But also think about more creative ways to solve this problem. (E.g., maybe you can mirror the phone screen to a desktop PC that can be monitored better by an observer? Or how about a screen recording that is analyzed later?) Decide on these limitations, write them down (see comments under “To deliver” in the box below) and then go ahead and do the experiment.

To deliver: Complete the study design in your PDF by a paragraph or subsection entitled **Limitations** that introduces the above decisions and compromises that you made. Shortly describe their relevance and how they might impact the result.

Some related comments and grading criteria: It is important that you clearly list the potential limitations that your compromises might have on your results. You can illustrate that also by stating how you did it here and how you would do it ideally. For example, something like: “For the sake of the test run, typing speed is measured with a stop watch by the experimenter. In the final experiment, it will be implemented on the phone.” Notice that in this example, you also need to clearly specify how “typing speed” is defined.

Fourth and final step: *Analyzing the results.* After finishing your tests, go and analyze the results. Because you will likely not have enough data anyhow and we have not discussed it in the lecture yet, you do not have to do a full statistical analysis (e.g., a significance test). Yet, depending on your project and what you measured, you should report, for example, averages, variances, etc. of your data.

There are no rules about what to analyze, because you specified that yourself already in the preceding steps (if you defined your research questions properly and came up with a correct study design). Likewise, there are no requirements for how to represent your results other than that it should make sense, answer your research question, and solve the problem you are studying. Think about what will be the best way to do this. Is it, for example, a table with the data? A diagram/chart? Both? If a diagram, what kind and how to plot the data? And do not forget to provide data about the subjects if you gathered it and if it is relevant.

Finally, do not just represent your results but draw some conclusions from them. First and foremost, use them to answer your research question. Because your study is only a little pre-experiment or pilot study, also discuss the issues that you might change for the final experiment. For example, as flawed as it may be due to the limitations, it could be that your little pre-experiment already identifies a clear trend that suggests to reconsider not just single aspects of the experiment design, but major things - in the extreme case even your original research question and goals.

To deliver: Add the results and analysis to your PDF. Structure it in the following way (minor changes are okay if you think they allow you to represent your results and conclusions in a better way):

- Results (*the actual data*)
- Discussion (*the analysis*)
- Conclusion (*this could include comments on how to change the actual experiment because of the conclusions drawn from your pilot study*)
- AI Disclosure Statement (*briefly specify which AI tools (if any) you used to help you with prototyping and data analysis, and how exactly how and to what extent they were used.*)

Some related comments and grading criteria: As above, clarity, correctness, and suitability of the presentation, discussion, and conclusions that you draw are the major grading criteria. Other aspects, such as presentation style, relevance of the text (everything that is needed, but no irrelevant and unnecessary information), etc. apply as well.